Report Template
FYS-STK3155 - Project X

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In this project we investigate how Ordinary Least Squares (OLS), Ridge, and Lasso regression perform when fitting the Runge function. We further explore different optimization techniques for training these models, including gradient descent, momentum methods, and stochastic gradient descent. To assess model performance, we apply bootstrapping and cross-validation and analyze the results in terms of the bias–variance trade-off.

I. INTRODUCTION

The problem of fitting models to data in order to understand its underlying distribution is one of the fundamental principles of statistics. The motivation stems from the benefit of predicting future outcomes based on past observations.

Ordinary Least Squares (OLS) regression provides a simple and widely used way to fit a model by finding the set of coefficients that minimizes the error with respect to a response variable y. While the model can capture both linear and non-linear relationships through basis expansions, it remains linear in its coefficients.

In many cases, however, some coefficients contribute little to explaining the variation in y. To address this, regularized versions of OLS have been developed. Ridge regression penalizes large coefficients to reduce variance, while Lasso regression introduces sparsity by shrinking some coefficients to zero, thereby performing variable selection.

Even when we obtain a seemingly good fit from these models, it is important to evaluate how robust the results are. Resampling techniques such as bootstrapping and cross-validation provide a way to assess the stability of the model and to estimate its predictive performance on unseen data.

In this project we will apply OLS, Ridge, and Lasso regression to the Runge function and investigate their performance under different levels of model complexity. We further explore optimization methods, including gradient descent, stochastic gradient descent, and momentum approaches, as alternatives to closed-form solutions. Finally, we study the bias–variance trade-off using resampling methods to better understand the strengths and limitations of these regression techniques.

II. METHODS

A. Method 1/X

Im am now citing hastie et .al in the text [1].

Wow we also used [2] for a lot of cool stuff

I have used chatGPT for a variety of the exercises week 39. Much of it to get used to latex syntax get the plots in the right format for the report with style for font. I also used chatgpt to give me feedback and slightly adjust my abstract and introduction.

B. Implementation

- Explain how you implemented the methods and also say something about the structure of your algorithm and present very central parts of your code, not more than 10 lines
- You should plug in some calculations to demonstrate your code, such as selected runs used to validate and verify your results. A reader needs to understand that your code reproduces selected benchmarks and reproduces previous results, either numerical and/or well-known closed form expressions.

C. Use of AI tools

- Describe how AI tools like ChatGPT were used in the production of the code and report.

III. RESULTS AND DISCUSSION

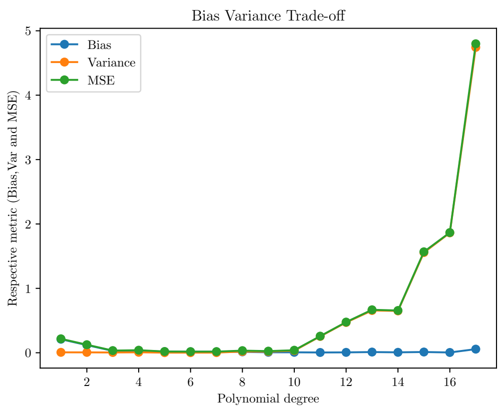

Figure 1: Bias Variance Tradeoff plot

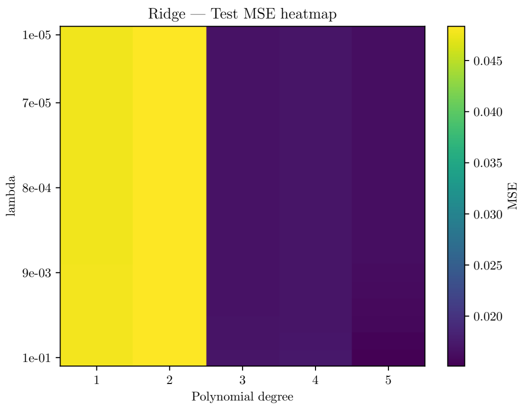

Figure 2: Heatmap of MSE for different penalty terms and polynomial degrees for ridge regression

- Present your results As we can see in Figure 1 i have done the proper reference. And i have also made a reference to the second image Figure 2. Wow
- Give a critical discussion of your work and place it in the correct context.
- Relate your work to other calculations/studies
- An eventual reader should be able to reproduce your calculations if she/he wants to do so. All input variables should be properly explained.
- Make sure that figures3 and tables contain enough information in their captions, axis labels etc. so that an eventual reader can gain a good impression of your work by studying figures and tables only.

Figure 3: My dog.

IV. CONCLUSION

- State your main findings and interpretations
- Try to discuss the pros and cons of the methods and possible improvements
- State limitations of the study
- Try as far as possible to present perspectives for future work

[1] T. Hastie, R.Tibshirani, and J.Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition. Springer Series in Statistics* (Springer, New York, 2009), URL `https://link.springer.com/book/10.1007%2F978-0-387-84858-7`.

[2] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., Journal of Machine Learning Research **12**, 2825 (2011), URL `https://scikit-learn.org/stable/about.html#citing-scikit-learn`.